

Gravity Lending

API Documentation for the Application Submission Endpoint

Doc Ver. 2.0



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Scope and Purpose

The scope of this document encompasses the API services provided by Gravity Lending as they relate to application submissions for Auto Refinance, Insurance, and Personal Loans. The purpose of this document is to help aid anyone who wishes to submit applications to Gravity Lending.

Prerequisites

Gravity Lending will provide:

- API Credentials (user, pass, and key)
- Source Code required in the body of your request
- Endpoint URL's for: Development, Staging, and Production

Endpoints accept only `POST` requests with a `JSON` request body by default.

By default all responses are `JSON` with `200 OK` HTTP code, otherwise an HTTP error code is returned, as explained later.

For the purposes of this document, only `JSON` will be used for examples.

Please speak with us about any specific XML needs you may have.

IMPORTANT!

You have been assigned a "Source Code".

A Source Code is used to attribute you as the originator of the application. It facilitates our ability to provide performance reporting for the traffic you send us.

Failure to pass the Source Code properly in the POST envelope will result in the new application being attributed to Gravity Lending.

Your Internal Unique Application Identifier:

If you have an id that identifies the application in your system. We will accept, and persist, it in our system through the `application.partner_loan_id`. as outlined in the Data Dictionary below. We will include it in the reporting that we provide back and it can potentially be used to aid in troubleshooting.

Hard Errors

Hard errors are those that return an HTTP response code other than 200 OK. The following is a list of hard error codes that can be returned by our API:

Client codes

Code	Description
400	The syntax is invalid. It is not properly formatted JSON or the nodes do not contain the proper keys.
401	An invalid <code>name/pass/apikey</code> was supplied in the <code>user</code> property.
403	The <code>name/pass/apikey</code> supplied is not authorized for the specified route.
404	The specified route does not exist.
405	An invalid method was requested (like trying to <code>PUT/GET/DELETE</code>), only <code>POST</code> is supported.
409	You are requesting an API version via the <code>X-Gravity-API-Version</code> header that is either invalid or no longer available.
413	<code>POST</code> data was too large. Limit is 2MB.

Server codes

Code	Description
500	An error happened in our API, please contact us immediately.
503	This could be a legit server response, please contact us immediately.

Please note that this response code list does not include normal responses that might get returned (Ex: 408 if there's a communication error), if there is a response code you receive that is not listed, please contact us and we can investigate.

Soft Errors

If you get a 200 OK you will be returned a JSON payload with a `success` and `error` key/value pair. If `success` is `false`, the error value will be populated with the error description. A soft error can occur if the API endpoint detects invalid data within a JSON payload, or for another reason, such as a hard error that was detected and handled gracefully. For an example of what the return JSON payload looks like, you can see below in the **Return Value** section.

The following is a list of *possible* errors that can be returned in the error key by this API:

- JSON error: data size larger than 2MB
- ERROR: applicant first name not found
- ERROR: applicant first name cannot be empty
- ERROR: applicant last name not found

- ERROR: applicant last name cannot be empty
- ERROR: applicant SSN not found
- ERROR: applicant SSN cannot be empty
- ERROR: applicant DOB not found
- ERROR: applicant DOB cannot be empty
- ERROR: applicant primary phone not found
- ERROR: applicant primary phone cannot be empty
- ERROR: applicant address not found
- ERROR: applicant address cannot be empty
- ERROR: applicant city not found
- ERROR: applicant city cannot be empty
- ERROR: applicant state not found
- ERROR: applicant state cannot be empty
- ERROR: applicant zip not found
- ERROR: applicant zip cannot be empty
- ERROR: applicant years not found
- ERROR: applicant years cannot be empty
- ERROR: applicant months not found
- ERROR: applicant months cannot be empty
- ERROR: applicant employment type not found
- ERROR: applicant employment type cannot be empty
- ERROR: applicant employment income not found
- ERROR: applicant employment income cannot be empty
- ERROR: vehicle year not found (N/A for personal loans)
- ERROR: vehicle year cannot be empty (N/A for personal loans)
- ERROR: vehicle make not found (N/A for personal loans)
- ERROR: vehicle make cannot be empty (N/A for personal loans)
- ERROR: vehicle model not found (N/A for personal loans)
- ERROR: vehicle model cannot be empty (N/A for personal loans)

This does not include potentially unhandled errors that could raise an exception. If an exception does occur, the exception error message (only the message and not the stack trace) will be returned as an error, please contact us if this happens.

Payload user Property

All endpoints require a `user` property in the root of the payload for access to be granted.

Example:

```
"user": {
  "name": "<assigned value>",
  "pass": "<assigned value>",
  "apikey": "<assigned value>"
}
```

Minimum Payload Example

All application submissions AUTO, INSURANCE, and PERSONAL require both the `user` and an `application` property to be considered valid.

(for personal loans, delete the `vehicle_*` properties from this sample)

```
{
  "user": {
    "name": "<assigned value>",
    "pass": "<assigned value>",
    "apikey": "assigned value"
  },
  "application": {
    "application_origin": "API",
    "source_code": "GRAVITY_DEV",
    "applicant_pp": false,
    "applicant_first_name": "La Toya",
    "applicant_last_name": "Jackson",
    "applicant_dob": "1956-05-29",
    "applicant_primary_phone": "8106505897",
    "applicant_email": "no-reply@gravitylending.com",
    "applicant_address": "116 Austin",
    "applicant_city": "Libertyville",
    "applicant_state": "IL",
    "applicant_zip": "60048",
    "applicant_years": 5,
    "applicant_months": 5,
    "applicant_employer_years": 5,
    "applicant_employer_months": 9,
    "applicant_employer_income": 11100.39,
    "applicant_employer_income_frequency": "Monthly",
    "applicant_employer_employment_type": "EMPLOYED",
    "vehicle_year": "2022",
    "vehicle_make": "Nissan",
    "vehicle_model": "Versa",
    "vehicle_trim": "SV Sedan 4D"
  }
}
```

Payload Data Dictionary

Note: If your intention is to include a co-applicant, data requirements are applied equally for both the primary and co applicants. As a co-applicant is NOT required neither are the co-applicant keys.

Node	Key	Req.	Type	Default	Description
user	name	YES	string	N/A	The user name trying to access this endpoint.
user	pass	YES	string	N/A	The password for the user trying to access this endpoint.
user	apikey	YES	string	N/A	The API key for the user trying to access this endpoint.
application	applicant_address	YES	string	N/A	Primary Applicant Street Address Concatenation of applicant_street_number and applicant_street_name
application	applicant_city	YES	string	N/A	Primary Applicant City
application	applicant_dob	YES	date	N/A	Primary Applicant Date of Birth (YYYY-MM-DD)
application	applicant_dl	NO	string	N/A	Primary Applicant Drivers License ID
application	applicant_dl_issue_date	NO	date	N/A	Primary Applicant Drivers License Issue Date (YYYY-MM-DD)
application	applicant_dl_expiration	NO	date	N/A	Primary Applicant Drivers License Expiration Date (YYYY-MM-DD)
application	applicant_email	YES	string	N/A	Primary Applicant Email Address
application	applicant_employer_employer	YES	string	N/A	Primary Applicant Employer
application	applicant_employer_employment_type	YES	ENUM	N/A	In Set [AT HOME, DISABILITY, EMPLOYED, EXECUTIVE, LABOURER, MANAGEMENT, MILITARY, OFFICE STAFF, OTHER, PRODUCTION, PROFESSIONAL, RETIRED, RETIRED - MILITARY, SALES, SELF - EMPLOYED, SIMI PROFESSIONAL, SERVICE, SOCIAL SECURITY, STUDENT, TRADES, UNEMPLOYED]

application	applicant_employer_income	YES	float	N/A	Primary Applicant Employer Income
application	applicant_employer_income_frequency	NO	ENUM	N/A	In Set [Weekly, Bi-Weekly, Monthly, Annually]
application	applicant_employer_months	NO	integer	N/A	Primary Applicant Employer For Months
application	applicant_employer_occupation	NO	string	N/A	Primary Applicant Employer Occupation
application	applicant_employer_years	NO	integer	N/A	Primary Applicant Employer For Years
application	applicant_first_name	YES	string	N/A	Primary Applicant First Name
application	applicant_language	NO	ENUM	N/A	In Set [ENGLISH, SPANISH]
application	applicant_last_name	YES	string	N/A	Primary Applicant Last Name
application	applicant_middle_name	NO	string	N/A	Primary Applicant Middle Name
application	applicant_months	YES	integer	N/A	Primary Applicant at Current Residence For Months
application	applicant_own_rent	YES	ENUM	N/A	In Set [OWN, RENT, FAMILY, OTHER]
application	applicant_payment	YES	float	N/A	Primary Applicant Housing Payment
application	applicant_pp	NO	boolean	0	Whether or not the applicant has granted permission for pulling credit
application	applicant_previous_address	NO	string	N/A	Primary Applicant Previous Street Address Concatenation of previous_applicant_street_number and previous_applicant_street_name
application	applicant_previous_city	NO	string	N/A	Primary Applicant Previous City
application	applicant_previous_employer_employer	NO	string	N/A	Primary Applicant Previous Employer
application	applicant_previous_employer_employment_type	NO	ENUM	N/A	In Set [AT HOME, DISABILITY, EMPLOYED, EXECUTIVE, LABOURER, MANAGEMENT, MILITARY, OFFICE STAFF, OTHER, PRODUCTION, PROFESSIONAL, RETIRED, RETIRED - MILITARY, SALES, SELF - EMPLOYED,

					SIMI PROFESSIONAL, SERVICE, SOCIAL SECURITY, STUDENT, TRADES, UNEMPLOYED]
application	applicant_previous_employer_months	NO	integer	N/A	Primary Applicant Previous Employer For Months
application	applicant_previous_employer_occupation	NO	string	N/A	Primary Applicant Previous Occupation
application	applicant_previous_employer_work_phone	NO	number	N/A	Primary Applicant Previous Employer Phone
application	applicant_previous_employer_years	NO	integer	N/A	Primary Applicant Previous Employer For Years
application	applicant_previous_months	NO	integer	N/A	Primary Applicant Previous Residence For Months
application	applicant_previous_own_rent	NO	ENUM	N/A	In Set [OWN, RENT, FAMILY, OTHER]
application	applicant_previous_street_name	NO	string	N/A	Primary Applicant Previous Street Name
application	applicant_previous_street_number	NO	string	N/A	Primary Applicant Previous Street Number
application	applicant_previous_suite_apartment	NO	string	N/A	Primary Applicant Previous Suite Apartment
application	applicant_previous_years	NO	integer	N/A	Primary Applicant Previous Residence For Years
application	applicant_previous_zip	NO	string	N/A	Primary Applicant Previous Zip Code
application	applicant_primary_phone	YES	number	N/A	Primary Applicant Phone Number (Digits ONLY)
application	applicant_secondary_phone	NO	number	N/A	Primary Applicant Secondary Phone Number (Digits ONLY)
application	applicant_ssn	NO	number	N/A	Primary Applicant Social Security Number (Digits Only)
application	applicant_state	YES	string	N/A	Primary Applicant State (2 characters)
application	applicant_street_name	NO	string	N/A	Primary Applicant Street Name
application	applicant_street_number	NO	string	N/A	Primary Applicant Street Number
application	applicant_suffix	NO	string	N/A	Primary Applicant Name Suffix
application	applicant_prefix	NO	string	N/A	Primary Applicant Name Prefix
application	applicant_suite_apartment	NO	string	N/A	Primary Applicant Suite Apartment

application	applicant_years	YES	integer	N/A	Primary Applicant at Current Residence For Years
application	applicant_zip	YES	string	N/A	Primary Applicant Zip Code
application	client_data_browser_fingerprint	NO	string	N/A	client browser type Landing Page Submissions
application	client_data_origination_ip	NO	string	N/A	client IP Address Landing Page Submissions
application	co_applicant_active	NO	boolean	0	Whether or not there is a CoApplicant to be considered with this application
application	co_applicant_address	NO	string	N/A	Co-Applicant Current Street Address Concatenation of co_applicant_street_number and co_applicant_street_name
application	co_applicant_city	NO	string	N/A	Co-Applicant Current City
application	co_applicant_dob	NO	date	N/A	Co-Applicant Date Of Birth (YYYY-MM-DD)
application	co_applicant_dl	NO	string	N/A	Co-Applicant Drivers License ID
application	co_applicant_dl_issue_date	NO	date	N/A	Co-Applicant Drivers License Issue Date (YYYY-MM-DD)
application	co_applicant_dl_expiration	NO	date	N/A	Co-Applicant Drivers License Expiration Date (YYYY-MM-DD)
application	co_applicant_email	NO	string	N/A	Co-Applicant Email Address
application	co_applicant_employer_employer	NO	string	N/A	Co-Applicant Current Employer
application	co_applicant_employer_employment_type	NO	ENUM	N/A	In Set [AT HOME, DISABILITY, EMPLOYED, EXECUTIVE, LABOURER, MANAGEMENT, MILITARY, OFFICE STAFF, OTHER, PRODUCTION, PROFESSIONAL, RETIRED, RETIRED - MILITARY, SALES, SELF - EMPLOYED, SIMI PROFESSIONAL, SERVICE, SOCIAL SECURITY, STUDENT,

					TRADES, UNEMPLOYED]
application	co_applicant_employer_income	NO	float	N/A	Co-Applicant Current Income
application	co_applicant_employer_income_frequency	NO	ENUM	N/A	In Set [Weekly, Bi-Weekly, Monthly, Annually]
application	co_applicant_employer_months	NO	integer	N/A	Co-Applicant Current Employer For Months
application	co_applicant_employer_occupation	NO	string	N/A	Co-Applicant Current Occupation
application	co_applicant_employer_years	NO	integer	N/A	Co-Applicant Current Employer For Years
application	co_applicant_first_name	NO	string	N/A	Co-Applicant First Name
application	co_applicant_language	NO	ENUM	N/A	In Set [ENGLISH, SPANISH]
application	co_applicant_last_name	NO	string	N/A	Co-Applicant Last Name
application	co_applicant_middle_name	NO	string	N/A	Co-Applicant Middle Name
application	co_applicant_months	NO	integer	N/A	Co-Applicant Current Residence For Months
application	co_applicant_own_rent	NO	ENUM	N/A	In Set [OWN, RENT, FAMILY, OTHER]
application	co_applicant_payment	NO	float	N/A	Co-Applicant Current Residence Payment
application	co_applicant_pp	NO	boolean	0	Whether or not the CO-applicant has granted permission for pulling credit
application	co_applicant_previous_address	NO	string	N/A	Co-Applicant Previous Street Address Concatenation of co_applicant_previous_street_number and co_applicant_previous_street_name
application	co_applicant_previous_city	NO	string	N/A	Co-Applicant Previous Residence City
application	co_applicant_previous_employer_employer	NO	string	N/A	Co-Applicant Previous Employer
application	co_applicant_previous_employer_employment_type	NO	ENUM	N/A	In Set [AT HOME, DISABILITY, EMPLOYED, EXECUTIVE, LABOURER, MANAGEMENT, MILITARY, OFFICE STAFF, OTHER, PRODUCTION, PROFESSIONAL, RETIRED, RETIRED - MILITARY, SALES, SELF - EMPLOYED,

					SIMI PROFESSIONAL, SERVICE, SOCIAL SECURITY, STUDENT, TRADES, UNEMPLOYED]
application	co_applicant_previous_employer_months	NO	integer	N/A	Co-Applicant Previous Employer For Months
application	co_applicant_previous_employer_occupation	NO	string	N/A	Co-Applicant Previous Occupation
application	co_applicant_previous_employer_work_phone	NO	number	N/A	Co-Applicant Previous Employer Phone (Digits Only)
application	co_applicant_previous_employer_years	NO	integer	N/A	Co-Applicant Previous Employer For Years
application	co_applicant_previous_months	NO	integer	N/A	Co-Applicant Previous Residence For Months
application	co_applicant_previous_own_rent	NO	ENUM	N/A	In Set [OWN, RENT, FAMILY, OTHER]
application	co_applicant_previous_street_name	NO	string	N/A	Co-Applicant Previous Residence Street Name
application	co_applicant_previous_street_number	NO	string	N/A	Co-Applicant Previous Residence Street Number
application	co_applicant_previous_suite_apartment	NO	string	N/A	Co-Applicant Previous Residence Suite Apartment
application	co_applicant_previous_years	NO	integer	N/A	Co-Applicant Previous Residence For Years
application	co_applicant_previous_zip	NO	string	N/A	Co-Applicant Previous Residence Zip Code
application	co_applicant_primary_phone	NO	number	N/A	Co-Applicant Phone Number (Digits Only)
application	co_applicant_secondary_phone	NO	number	N/A	Co-Applicant Secondary Phone Number (Digits Only)
application	co_applicant_ssn	NO	number	N/A	Co-Applicant Social Security Number (Digits Only)
application	co_applicant_state	NO	string	N/A	Co-Applicant Current State
application	co_applicant_street_name	NO	string	N/A	Co-Applicant Current Street Name
application	co_applicant_street_number	NO	string	N/A	Co-Applicant Current Street Number
application	co_applicant_suffix	NO	string	N/A	Co-Applicant Name Suffix
application	co_applicant_prefix	NO	string	N/A	Co-Applicant Name Prefix
application	co_applicant_suite_apartment	NO	string	N/A	Co-Applicant Current Suite Apartment
application	co_applicant_years	NO	integer	N/A	Co-Applicant Current Residence For Years
application	co_applicant_zip	NO	string	N/A	Co-Applicant Current Zip Code

application	marketing_intent	NO	string	N/A	Applicant Indicated motive to applying
application	partner_loan_id	NO	string	N/A	dedicated field for tracking the sources internal unique application identifier
application	marketing_sub_ids	NO	object	N/A	{"source_id": "Open for your use string", "MarketingSubID1": "", "MarketingSubID2": "", "MarketingSubID3": "", "MarketingSubID4": ""}
application	partner_loan_id	NO	string	N/A	Source Partners Internal Application UUID
application	quotes_customer_reported_next_payment_date	NO	date	N/A	Current Loan Next Payment Due Date
application	quotes_customer_reported_payment	NO	float	N/A	Current Loan Payment
application	quotes_customer_reported_payoff	NO	float	N/A	Current Loan Payoff Amount
application	quotes_customer_reported_rate	NO	float	N/A	Current Loan Interest Rate
application	source_code	YES	string	GRAVL END	Assigned by Gravity Lending
application	vehicle_make	YES	string	N/A	The Vehicle Manufacture
application	vehicle_miles	YES	integer	N/A	The Vehicle Odometer Reading
application	vehicle_model	YES	string	N/A	The Vehicle Model
application	vehicle_trim	NO	string	N/A	The Vehicle Trim Description
application	vehicle_vin	NO	string	N/A	17 Character Vehicle Identification
application	vehicle_year	YES	number	N/A	Four Digit Vehicle Year

Return Value

The response is a JSON string in the following format

Example:

```
{
  "success": true/false,
  "data": "requested data",
  "error": "error string"
}
```

If there is no error, `success` will be `true` and the `error` value will be empty, otherwise `success` will be `false` and the `error` value will be the error string pertaining to any issues that occurred within the API upon application submission.

It is possible that a `data` node will also be included in the return value. If there is a `data` node, this node may contain additional data as it relates to the request.

When an application is submitted, the `data` node will contain the new Gravity Lending application ID. Additionally, if there are errors when trying to pull credit or get the vehicle valuations, the `data` node will also contain these errors.

It should be noted that if the application successfully submits, but there are bureau or vehicle valuation errors, `success` will still be `true`, and the error string will still be empty. This is because the application was submitted and thus is in our system, and those errors are 3rd party errors that can be corrected manually.

Example:

```
{
  "success": true,
  "data": {
    "application_id": 74,
    "nada_error": "No data available for the input vehicle parameters in the input period/date",
    "bureau_errors": [ "Error Code: 02B, CERR - Processing: CA - CITY OR STATE IS MISSPELLED OR NOT IN ACROPAC SYSTEM" ]
  },
  "error": ""
}
```

Returning Offers

This endpoint can also return “offers” that Gravity Lending has determined to be a good match based on certain factors of the application.

These factors include the applicant’s FICO score, the vehicle year of the car (if the application is an auto application), the applicant’s state and the amount being requested for the loan.

We use this information to return possible offers that the applicant might qualify for based on rates that our lender partners have provided to us. These offers are internal to Gravity Lending only: We do not send the application to any lender partner to retrieve these rates as we maintain these rate sheets internally.

Please note in the rare instance that credit data cannot be pulled, we use a default FICO value of 680. This covers most of our lender partners and allows for potential offers to still be returned; in this instance, if we are later able to successfully pull the applicant’s FICO, we can then adjust the offers based on the pulled FICO value.

An example payload with offers may look like the following:

```
{
  "success": true,
  "data": {
    "application_id": 280,
    "offers": {
      "GravityOffers": {
        "offer-1": {
          "Offerid": 1,
          "LoanAmount": "6026",
          "InterestRate": "5.64",
          "Term": "84",
          "MonthlyPayment": "86",
          "APR": "5.64"
        },
        "offer-2": {
          "Offerid": 2,
          "LoanAmount": "6026",
          "InterestRate": "4.24",
          "Term": "78",
          "MonthlyPayment": "87",
          "APR": "4.24"
        }
      }
    }
  },
  "error": ""
}
```

The number of offers returned will vary. If no offers are returned, this likely means that we do not have any lenders who operate in the state of the applicant.

Developer Notes

If you are including a co-applicant, all data requirements are identical to that of the main applicant.

Use caution when making changes involving applicant properties. Double check that your changes are always paired between the primary and co-applicant keys.

The key names for BOTH applicants are identical except for the first two characters. e.g. “applicant” is the primary applicant and “co_applicant” is the co-applicant.

The Data Dictionary contains all the details you will need. Please consult it first, and often, to aid your journey. If you have other data not defined there, that would further our shared goals, please let us know.

Start with the MINIMUM example included above

Add values in pairs to keep parity between the primary and co applicant data.

CLI cURL:

```
curl -X POST -i https://wf.gravitylending.dev:8080/applications/submit -d '{
  "user": {
    "name": "GRAVITY_DEV",
    "pass": "REDACTED",
    "apikey": "REDACTED"
  },
  "options": {
    "application_type": "AUTO"
  },
  "application": {
    "application_origin": "API",
    "source_code": "GRAVITY_DEV",
    "applicant_pp": "false",
    "applicant_first_name": "Latoya",
    "applicant_last_name": "Jackson",
    "applicant_dob": "1973-12-01",
    "applicant_primary_phone": "8106505897",
    "applicant_email": "no-reply@gravitylending.com",
    "applicant_address": "116 Austin",
    "applicant_city": "Libertyville",
    "applicant_state": "IL",
    "applicant_zip": "60048",
    "applicant_years": 5,
    "applicant_months": 5,
    "applicant_employer_employer": "Source Integrations",
    "applicant_employer_years": 5,
```



```
"applicant_employer_months": 9,  
"applicant_employer_income": 11100.39,  
"applicant_employer_income_frequency": "Monthly",  
"applicant_employer_employment_type": "EMPLOYED",  
"co_applicant_active": false,  
"vehicle_year": "2022",  
"vehicle_make": "Nissan",  
"vehicle_model": "Versa",  
"vehicle_trim": "SV Sedan 4D"  
}  
'
```

Expected SUCCESS

```
HTTP/1.1 200 OK  
Date: Thu, 01 Apr 2021 20:35:34 GMT  
Server: -  
Content-Security-Policy: default-src https: data: 'unsafe-inline'  
'unsafe-eval'  
Strict-Transport-Security: max-age=31536000; includeSubDomains  
X-Xss-Protection: 1; mode=block  
X-Content-Type-Options: nosniff  
Transfer-Encoding: chunked  
Content-Type: application/json  
  
{"success":true,"data":{"application_id":171320},"error":""}
```

Expected FAILURE:

```
HTTP/1.1 200 OK  
Date: Thu, 01 Apr 2021 20:35:34 GMT  
Server: -  
Content-Security-Policy: default-src https: data: 'unsafe-inline'  
'unsafe-eval'  
Strict-Transport-Security: max-age=31536000; includeSubDomains  
X-Xss-Protection: 1; mode=block  
X-Content-Type-Options: nosniff  
Transfer-Encoding: chunked  
Content-Type: application/json  
  
{"success":false,"error":"JSON error: applicant first name not found"}
```